THE LANGUAGE OF

MATHEMATICS

General notation

Symbol	Meaning	Examples
$=$	Equal	$2+3=5$; $4-4=0$
$\neq$	Not equal	$2\neq 3$; $-1\neq 1$
$\approx$	Approximately	$\pi\approx 3.14$ $\sqrt{2}\approx 1.41$
$<,\ >$	Less than; greater than	$2<3$; $-1<0$ $5>4$; $0>-2$
$\leq,\ \geq$	Less than or equal to; greater than or equal to	$2\leq 2$; $-1\leq 3$ $5\geq 4$; $0\geq 0$
$\pm$	Plus or minus	$3\pm 1=4$ or 2 $-2\pm 3=1$ or -5
$\lvert a\rvert$	Absolute value	$\lvert -5\rvert=5$; $\lvert 3\rvert=3$
$\sqrt{a}$	Square root	$\sqrt{9}=3$; $\sqrt{\frac{1}{4}}=\frac{1}{2}$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set with three elements	$\{2,4,6\}$; $\{-1,0,5\}$
$\varnothing$	Empty set	$\{1\}\cap\{2\}=\varnothing$ $\{x\in\mathbb{R}: x^2=-1\}=\varnothing$
$\in$	Is element of	$2\in\{1,2\}$; $-1\in\mathbb{Z}$
$\notin$	Is not element of	$3\notin\{1,2\}$; $\frac{1}{2}\notin\mathbb{Z}$
$\subseteq$	Is a subset of	$\{1\}\subseteq\{1,2\}$ $\mathbb{N}\subseteq\mathbb{Z}$
$\cup,\ \cap$	Union; intersection	$\{1\}\cup\{2\}=\{1,2\}$ $\varnothing\cup\{0\}=\{0\}$ $\{1,2\}\cap\{2\}=\{2\}$ $\{-1\}\cap\{4\}=\varnothing$
$\mathbb{N},\ \mathbb{Z}$	Natural numbers; integers	$3\in\mathbb{N}$; $8\in\mathbb{N}$ $-2\in\mathbb{Z}$; $0\in\mathbb{Z}$
$\mathbb{Q},\ \mathbb{R}$	Rational numbers; real numbers	$\frac{1}{2}\in\mathbb{Q}$; $-3\in\mathbb{Q}$ $\sqrt{2}\in\mathbb{R}$; $\frac{1}{4}\in\mathbb{R}$
$(a,b),\ [a,b]$	Open interval; closed interval	$1\notin(1,3)$; $0\in(-2,2)$ $1\in[1,3]$; $-2\in[-2,0]$

In the implications below, $x\in\mathbb{R}$.

Logic

Symbol	Meaning	Examples
$\Rightarrow$	Implies	$x>2\Rightarrow x>1$ $x=3\Rightarrow x^2=9$
$\Leftrightarrow$	If and only if	$x+1=3\Leftrightarrow x=2$ $x^2=0\Leftrightarrow x=0$
$\forall$	For every	$\forall x\in\mathbb{R},\ x^2\geq 0$ $\forall n\in\mathbb{Z},\ n+1\in\mathbb{Z}$
$\exists$	There exists	$\exists x\in\mathbb{Z}:\ x+2=0$ $\exists t\in\mathbb{R}:\ t^2=2$
$\neg$	Logical negation	$\neg(2>3)$ is true $\neg(4\in\{1,2\})$ is true

Summation notation

Symbol	Meaning	Examples
$\sum_{i=1}^{n} a_i$	Add a_i for each integer i from 1 to n.	$\sum_{i=1}^{3} 2i=2+4+6$ $\sum_{i=1}^{2}(i+3)=4+5$

In $\sum_{i=1}^{n} a_i$, i is the **index**, 1 is the **lower bound**, n is the **upper bound**, and a_i is the **expression being summed**.

Worked examples

$\sum_{i=1}^{4} i=1+2+3+4=10$

$\sum_{k=1}^{3} k^2=1+4+9=14$

Symbols acquire meaning from their mathematical context.

Alternatives: fair die (I) or fair card draw (II).
I: $A=\{1,3,5\}$, $B=\{1,2\}$.
II: cards 1–4; $A=\{4\}$, $B=\{2,3,4\}$.

Probability and events

Symbol	Meaning	Examples
Ω	Sample space: all possible outcomes	I: $\Omega=\{1,\ldots,6\}$ II: $\Omega=\{1,2,3,4\}$
ω	An individual outcome	I: $\omega=3\in\Omega$ II: $\omega=4\in\Omega$
$A=\{1,3,5\}$	Event shown as a set of outcomes	I: $A=\{1,3,5\}$: odd II: $A=\{4\}$: card 4
$A\subseteq\Omega$	An event is a subset of the sample space	I: $\{1,3,5\}\subseteq\Omega$ II: $\{4\}\subseteq\Omega$
$P(A)$	Probability of event A	I: $P(A)=\frac{3}{6}=\frac{1}{2}$ II: $P(A)=\frac{1}{4}$
A^c	Complement of event A	I: $A^c=\{2,4,6\}$ II: $A^c=\{1,2,3\}$
$A\cup B$	Event A or B (or both)	I: $A\cup B=\{1,2,3,5\}$ II: $A\cup B=\{2,3,4\}$
$A\cap B$	Both events A and B	I: $A\cap B=\{1\}$ II: $A\cap B=\{4\}$
$P(A\mid B)$	Probability of A given B	I: $P(A\mid B)=\frac{1}{2}$ II: $P(A\mid B)=\frac{1}{3}$

Functions: I: $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^2$.
II: $f:[0,\infty)\to\mathbb{R}$, $f(x)=\sqrt{x}$.

Function notation

Symbol	Meaning	Examples
f	Name of a function	I: $f(x)=x^2$ II: $f(x)=\sqrt{x}$
$f(x)$	Value of f at input x	I: $f(3)=3^2=9$ II: $f(4)=\sqrt{4}=2$
$f:A\to B$	Function from set A to set B	I: $f:\mathbb{R}\to\mathbb{R}$ II: $f:[0,\infty)\to\mathbb{R}$
$x\mapsto f(x)$	Input-to-output mapping	I: $x\mapsto x^2$; $3\mapsto 9$ II: $x\mapsto\sqrt{x}$; $4\mapsto 2$
$\mathrm{dom}(f)$	Domain of a function	I: $\mathrm{dom}(f)=\mathbb{R}$ II: $\mathrm{dom}(f)=[0,\infty)$

Below: triangle angles (α,β,γ); $\Theta=2\theta$;
$\Delta x=x_2-x_1$; population mean μ and SD σ.

Letters and Greek symbols

Symbol	Meaning	Examples
$a,b,c,\ldots$	Common lowercase Latin letters	$a=2,\ b=3,\ c=5$ $a=-1,\ b=4,\ c=3$
$A,B,C,\ldots$	Common uppercase Latin letters	Here $C=A\cup B$: $A=\{1\},\ B=\{2\}$ $A=\{0,3\},\ B=\{3\}$
x,y,z	Other common Latin letters	$x=2,\ y=4,\ z=6$ $x=-1,\ y=1,\ z=-2$
$\alpha,\ \beta,\ \gamma$	Alpha; beta; gamma	$(60^\circ,60^\circ,60^\circ)$ $(30^\circ,70^\circ,80^\circ)$
$\delta,\ \Delta$	Delta; uppercase delta	$\delta=0.1$; $\delta=0.05$ $\Delta x=5-2=3$ $\Delta x=1-4=-3$
$\theta,\ \Theta$	Theta; uppercase theta	$\theta=30^\circ$; $\Theta=60^\circ$ $\theta=45^\circ$; $\Theta=90^\circ$
$\lambda,\ \Lambda$	Lambda; uppercase lambda	$\lambda=2$; $\Lambda=\{1,2\}$ $\lambda=-1$; $\Lambda=\{0,3,5\}$
$\mu,\ \sigma,\ \Sigma$	Mu; sigma; uppercase sigma	$\{3,7\}:\ \mu=5,\ \sigma=2$ $\{2,10\}:\ \mu=6,\ \sigma=4$ $\sum_{i=1}^{3} i=6$; $\sum_{k=1}^{2} k^2=5$
$\pi,\ \Pi$	Pi; uppercase pi	$\pi\approx 3.14$ (constant) $\prod_{j=1}^{3} j=6$ $\prod_{k=1}^{2}(k+2)=12$
$\omega,\ \Omega$	Omega; uppercase omega	$\Omega=\{0,1\},\ \omega=0$ $\Omega=\{1,\ldots,6\},\ \omega=5$